

Yujie Zang

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EDUCATION

National University of Singapore

Singapore

M.Sc. in Robotics

Jul. 2026 – Present

Selected Coursework: Robot Kinematics, Robot Dynamics and Control, AI Planning and Decision Making, Machine Learning in Robotics

Shanghai Jiao Tong University

Shanghai, China

B.Eng. in Mechanical Industrial Design (Outstanding Graduate)

Aug. 2022 – Jun. 2026

Grades: 90.10 / 100 | GPA: 3.91 / 4.30 | Ranking: 4 / 36

Selected Coursework: Product Design, Materials and Structures, Advanced Mathematics, Computer Programming, Data Science, Machine Learning

RESEARCH INTEREST

My long-term research goal is to build robotic systems that leverage **human priors** to advance toward **superhuman physical intelligence**. In particular, I focus on **real-world robotic manipulation**, using **force and tactile sensing** to understand physical interactions, **predictive model** to predict their evolution, and **real-robot post-training** to acquire precise and fast skills.

PUBLICATIONS

(† equal contribution; * corresponding author)

[1] TacForeSight: Force-Guided Tactile World Model for Contact-Rich Manipulation.

Yujie Zang†, Yuhang Zheng†, Xian Nie†, Yupeng Zheng, Shuai Tian, Songen Gu, Chen Gao, Zining Wang, Shuicheng Yan*, Wenchao Ding*.

IEEE Robotics and Automation Letters, 2026. (Under Review)

[2] Interaction Latents As Actions for Contact-Rich Manipulation.

Yuhang Zheng†, Yujie Zang†, Yuchen Liu†, Yupeng Zheng, Zhennan Jiang, Zebin Xing, Shuai Tian, Songen Gu, Jingxiang Guo, Chen Gao, Zhongpu Xia*, Shuicheng Yan*.

International Conference on Learning Representations, 2027. (Under Review)

[3] Contact Trajectory Prompting: In-Context Transfer of Contact-Rich Behaviors from a Single Demonstration.

Xian Nie†, Yujie Zang†, Yuhang Zheng†, Yupeng Zheng, Wendi Chen, Chuan Wen, Cewu Lu, Wenhao Ding, Junchi Yan*, Shuicheng Yan*.

IEEE International Conference on Robotics and Automation, 2027. (Under Review)

[4] Multimodal Palpation Sensing for Precise Prediction of Breast Lump Hardness and Size.

Yujie Zang, Shuhong Xiao, Peng Yang, Ying Yang, Zixuan Lu, Ruyue Sun, Chongpeng Wei, Shuyue Zheng, Chen Wang, Xinyang Tan*, Hongwei Zhang*.

IEEE International Conference on Acoustics, Speech and Signal Processing, 2026. (Accepted)

[5] Re-DirectExo: A Preliminary Study of Illusion-Driven Hand Rehabilitation Exoskeleton.

Yujie Zang, Guiqin Lu, Zhaoyi Chen, Zhen Han, Zihang Ma, Xinyang Tan*, Shi Qiu*.

International Conference on Human-Engaged Computing, 2025. (Accepted)

[6] GAEA-Net: Generating Activity-Enriched Abnormal ECGs via Adversarial Network.

Liuqing Chen, Shuhong Xiao, Yujie Zang, Jiner Wang, Shanhai Hu, Lin Zhang, Qing Li, Danyang Zhang.

IEEE Journal of Biomedical and Health Informatics, 2026. (Accepted)

[7] OmniVTA: Visuo-Tactile World Modeling for Contact-Rich Robot Manipulation. (EAI-100 Top 10 Opensouce Embodied AI Dataset, 2025)

Yuhang Zheng, Songen Gu, Weize Li, Yupeng Zheng, Yujie Zang, Shuai Tian, Xiang Li, Ruihai Wu, Ce Hao, Chen Gao, Si Liu, Haoran Li, Yilun Chen, Shuicheng Yan*, Wenchao Ding*.

IEEE Transactions on Robotics, 2026. (Under Review)

[8] **VT-WAM: Visual-Tactile World Action Model for Contact-Rich Manipulation.**
Shuai Tian, Yupeng Zheng, Yuhang Zheng, Songen Gu, **Yujie Zang**, Yuxing Qin, Weize Li, Haoran Li, Wenchao Ding*, Dongbin Zhao*.
IEEE Robotics and Automation Letters, 2026. (Under Review)

RESEARCH EXPERIENCE

LV-Robotics Lab, National University of Singapore	Singapore
Research Intern Advisor: Prof. Shuicheng Yan	Aug. 2026 – Present
Topics: Contact-rich manipulation; tactile sensing and robot learning	
Intelligent Medical Design Lab, Shanghai Jiao Tong University	Shanghai, China
Research Intern Advisors: Prof. Guangye Li, Prof. Xinyang Tan	Apr. 2023 – Jun. 2026
Topics: EEG decoding and wearable sleep monitoring product development; medical hardware devices (hand exoskeleton, tumour detection product, scoliosis orthoses design)	
Institute for AI Industry Research (AIR), Tsinghua University	Beijing, China
Research Intern Advisors: Prof. Wenchao Ding	Jan. 2025 – Mar. 2025
Topics: Precious Manipulation and Robot Understanding. Evaluated whether robotic agents understand complex 3D environments.	
D3 Lab, College of Computer Science and Technology, Zhejiang University	Hangzhou, China
Research Intern Advisor: Prof. Liuqing Chen	Nov. 2024 – Apr. 2025
Topics: Activity-enriched ECG synthesis; GAN modeling for ECG signals	

INDUSTRY EXPERIENCE

Romoya Robotics	Singapore
Research Assistant Advisor: Prof. Shuicheng Yan	Aug. 2026 – Present
Topics: Robot System Development, Robot Research Assistant	
Anyverse Dynamics (无界动力)	Beijing, China
Robotics Algorithm Research Intern Advisor: Prof. Zhongpu Xia	Jun. 2026 – Aug. 2026
Topics: Dual Arm Manipulation Pipeline, Human-in-the-Loop Robot Learning, Real Robot Reinforcement Learning	
TARS Robotics (它石智航)	Shanghai, China
Robotics Algorithm Research Intern Advisor: Prof. Wenchao Ding	Jan. 2026 – Jun. 2026
Topics: Tactile Manipulation, Robot Learning, Tactile UMI Hardware Design	

AWARDS

Champion, RoboMaster University Championship	Sep. 2024
Third Prize, China–U.S. Young Maker Competition (Shanghai Division)	Jul. 2024
Honorable Mention, Mathematical Contest in Modeling (MCM)	May 2024

SKILLS

Robotics & Hardware: Mechanical design (SolidWorks, CAD, machining and assembly), embedded electronics, PCB and sensor integration, ROS/ROS2, robot kinematics, real-robot deployment, and ergonomic prototyping.

Robot Learning & Programming: Reinforcement learning, imitation learning, diffusion policy, tactile manipulation, and human-in-the-loop learning. Python, MATLAB, C/C++, Git; Unity, Blender, LaTeX, and Figma.

Data Analysis & Languages: Signal processing (EEG, ECG, sEMG), machine learning, deep learning, time-series analysis, and data visualization. Mandarin (Native), English (Fluent).